home lab project, week 3 day 4

Task:

Test the complete lab environment against the project requirements, correct configuration issues, and document the results.

1. Review the project requirements

Project Requirement	What you should verify
Virtualized cybersecurity lab	Kali and Ubuntu are running correctly in UTM
Isolated network	Both VMs use the intended `192.168.64.0/24` network
Kali testing system	Kali is configured and has network connectivity
Ubuntu protected system	Ubuntu is configured and reachable from the lab
Network communication	Kali and Ubuntu can communicate
Controlled testing	Kali can perform authorized tests against Ubuntu
Packet analysis capability	Wireshark can be installed/configured on Kali
Recoverability	VM snapshots/baselines are available if required
Documentation	Configuration and test results are recorded

I am testing the **current Week 3 implementation against the requirements that should already be achievable**.

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2. Check the Kali configuration

On Kali, verify:

ip addr

Confirm that Kali has:

192.168.64.2

Then:

ip route

Confirm that the `192.168.64.0/24` network is present.

Then test Ubuntu:

ping -c 4 192.168.64.3

You want successful replies.

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3. Check the Ubuntu configuration

On Ubuntu:

ip addr

Confirm:

192.168.64.3

Then:

ip route

Confirm that Ubuntu has a route for:

192.168.64.0/24

Then test Kali:

ping -c 4 192.168.64.2

This verifies communication in both directions.

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4. Test host discovery from Kali

Since Kali is your testing system, perform a basic discovery test against your **own isolated lab network**:

nmap -sn 192.168.64.0/24

You should be able to identify the systems participating in your lab network.

For example, you should see your Ubuntu system at:

192.168.64.3

This is a good first validation that the network is functioning from the perspective of your cybersecurity testing system.

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5. Test whether the required security tools work

For example, check whether Wireshark is available:

wireshark --version

Or open Wireshark from the Kali application menu.

You don't need to perform your full packet-analysis exercise yet. You're simply verifying that the environment is ready for the upcoming project activities.

You can also verify that Nmap works:

nmap --version

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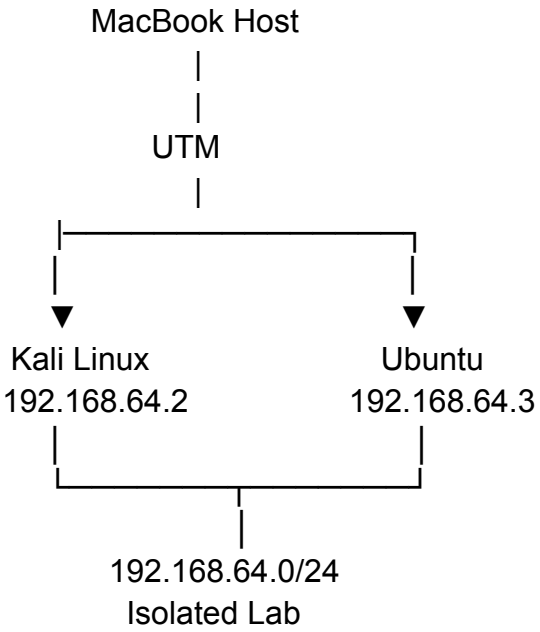
6. Check the VM/network configuration in UTM

This is important because your lab depends on the virtualization configuration.

For both Kali and Ubuntu, verify:

- VM starts normally
- Correct virtual network adapter is attached
- Correct network type is selected
- Expected IP address is assigned
- VM has the appropriate CPU/RAM/storage
- No accidental connection to an unintended network
- Snapshots/baselines are available if you created them

Your documentation should show the relationship:



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7. Look for configuration problems

This is where the **"correct configuration issues"** part of your task comes in.

Don't intentionally create problems just so you can say you fixed something.

Instead, check for actual problems.

Check:

ip addr
ip route

Then check the UTM virtual network configuration.

No issues in the network configuration, and Kali can communicate with Ubuntu, and Ubuntu can also communicate back.

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8. Create a requirements validation table

This is probably the **best evidence for Day 4**.

You can create something like this in your project documentation:

Requirement	Test Performed	Expected Result	Actual Result	Status
Kali VM operational	Started Kali VM	VM starts normally	Kali started successfully	PASS
Ubuntu VM operational	Started Ubuntu VM	VM starts normally	Ubuntu started successfully	PASS
Kali IP configuration	`ip addr`	`192.168.64.2`	`192.168.64.2`	PASS
Ubuntu IP configuration	`ip addr`	`192.168.64.3`	`192.168.64.3`	PASS
Virtual network connectivity	Kali ping Ubuntu	Successful replies	Successful replies	PASS
Reverse connectivity	Ubuntu ping Kali	Successful replies	Successful replies	PASS
Network discovery	`nmap -sn`	Ubuntu discovered	Ubuntu discovered	PASS
Wireshark availability	Launch/version test	Wireshark operational	Operational	PASS

Lab isolation	Review UTM network settings	Isolated lab network	Verified	PASS

Don't enter PASS until you've actually performed each test. If something fails, record it as FAIL, fix it, and then retest.

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9. Document any corrections

Coming across an issue is a good practice. I should document it, rather than hide it.

For example:

Issue	Correction	Retest Result
Ubuntu initially had incorrect network configuration	Corrected virtual network configuration	Connectivity successful
Kali could not reach Ubuntu	Corrected network adapter settings	Ping successful

If everything works on the first test, that's fine too. You can document:

> **No configuration issues were identified during validation. The lab environment met the current Week 3 requirements.**

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10. Capturing Evidence

For Day 4, I recommend collecting approximately **5–7 screenshots**.

Screenshot 1 — Kali configuration

ip addr

Shows:

192.168.64.2

Screenshot 2 — Ubuntu Configuration

ip addr

Shows:

192.168.64.3

Screenshot 3 — Kali → Ubuntu

ping -c 4 192.168.64.3

Screenshot 4 — Ubuntu → Kali

ping -c 4 192.168.64.2

Screenshot 5 — Network discovery

nmap -sn 192.168.64.0/24

Screenshot 6 — Wireshark

Show that Wireshark launches and your network interface is available.

Screenshot 7 — Requirements validation table

Show your completed PASS/FAIL validation table.

What Day 4 should accomplish

Think of Week 3 like this:

Day 1: Install Kali
↓
Day 2: Install Ubuntu
↓
Day 3: Connect and test Kali ↔ Ubuntu
↓
Day 4: Test the whole lab against the requirements

So Day 4 is essentially your **Week 3 acceptance/validation test**.